

# Utkarsh Jaiswal

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## Education

**Indian Institute of Technology, Kharagpur**

2018 – 2023

*Dual Degree (B.Tech + M.Tech) in Exploration Geophysics (GPA: 8.37 / 10)*

## Experience

**Indian Political Action Committee**

Jun 2023 – Present

*Full Stack Developer*

*Hyderabad*

- Designed and implemented a **real-time election dashboard** capable of processing voter data dynamically, enabling in-depth analysis across constituencies with metrics such as voter demographics, turnout percentages etc
- Designed and implemented modular components for a scalable data analytics platform, enabling seamless generation of accurate prediction reports from diverse datasets using **Python** and **Django**
- Integrated **AWS S3** within Django APIs to enable secure file uploads and retrieval, enhancing system functionality
- Enhanced API response times by implementing efficient data caching strategies like **Redis**, reducing database load

**Firstbench.AI**

Aug 2024 – Oct 2024

*Backend Developer Freelance*

- Led the backend development team, designing and implementing a user management system from scratch using **Node.js**, **TypeScript**, and **MongoDB**, integrating **Google OAuth** for secure login and **Twilio** for OTP verification
- Successfully deployed the application on **AWS EC2** and implemented Continuous Integration and Continuous Deployment (CI/CD) pipeline using **Docker** and **Jenkins**, streamlining code deployment and reducing downtime
- Built real-time debate module with timed speaking sessions, **WebRTC**, and **Socket.io** for multi-user discussions and integrated **Whisper API** for speech-to-text transcription and AI-driven response evaluation

**GurukulBytes**

Jan 2022 – Apr 2022

*Full Stack Developer Intern*

- Developed an interactive module using **Angular**, enabling users to solve coding problems with an intuitive interface
- Implemented a robust backend using **Scala** and **MongoDb**, ensuring efficient management of coding problem data
- Developed a cloud-based code evaluation system using Judge0 compiler on AWS EC2, accepting submissions in multiple programming languages, and providing real-time feedback on test case passing, errors, and TLE status
- Added functionality in admin portal to upload multiple coding questions and test cases from txt files at one go

## Projects

**AI-Powered Database Query App** | [Repository Link](#)

- Developed a user-friendly **Streamlit** app employing a Retrieval-Augmented Generation (RAG) approach with Gemini LLM and **LlamaIndex** to convert complex natural language user inputs into actionable SQL queries
- Integrated **VectorStoreIndex** leveraging similarity search and semantic **vector embeddings** for real-time retrieval of database metadata from PostgreSQL and MySQL, enabling fast, context-aware query generation

**Cloud Notebook** | [Repository Link](#)

- Architected a secure **MERN** stack note-making web-app with authentication middleware, ensuring users can only access and modify their own notes by implementing **RESTful** APIs using **MongoDB** for efficient data persistence

**News Application** | [Repository Link](#)

- Built a responsive news application using **ReactJs** integrating real-time updates from [newsapi.org](https://newsapi.org/)
- Implemented infinite scrolling and category-based filtering (sports, entertainment etc) to enhance user experience

**Felvin Search** | [Open Source Contribution](#)

- Contributed to the Instant Apps repository by developing interactive cards for prime number verification and numeral conversion by implementing efficient algorithm to ensure optimal performance of the features

## Technical Skills and Achievements

**Technologies:** Python, C/C++, JavaScript, TypeScript, Django, Node.js, Express.js, MongoDB, AWS, Docker, Kubernetes, Linux, Git/Github, Jenkins

**Competitive Programming:** Secured highest **Codeforces** rating of **1612** and solved more than 1500 programming questions on platforms like **Codeforces**, **Leetcode**, **InterviewBit**